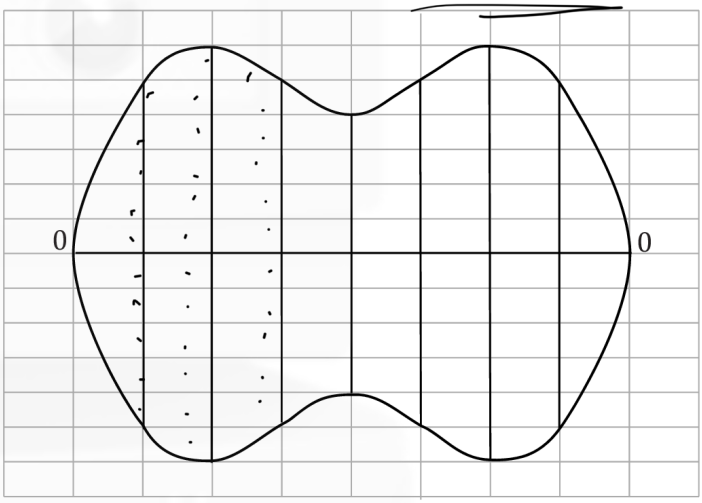
Question 5 **(25 marks)**

Harry draws a scale diagram of the portion of his garden that is covered in lawn.
His diagram is shown below.
Each box on the grid is 1 cm × 0·5 cm.
Each cm on Harry's diagram represents 3 m.
In order to estimate the area of the lawn Harry divides the diagram into eight sections.

(a) Use the trapezoidal rule to estimate the area of the lawn using the scale: 1 cm = 3 m.



$$A = \frac{h}{2}(y_1 + y_n + 2(y_2 + y_3 + y_4 + \ldots + y_{n-1}))$$

$$A = \frac{3}{2}(0 + 0 + 2(15 + 18 + 15 + 12 + 15 + 18 + 15))$$

$$A = \frac{3}{2}(0 + 0 + 216)$$

$$A = 324 \text{ m}^2$$

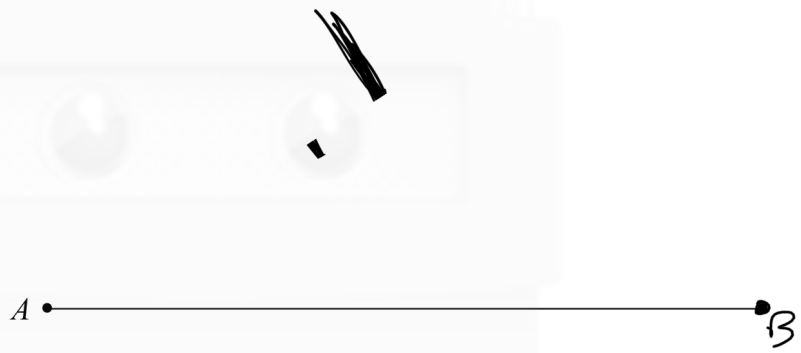
(b) Nuala can walk at a speed of 1·6 metres per second.
Write this speed in kilometres per hour.

$1 \cdot 6 \times 60 \times 60 \div 1000 = 5.76$

Marking Scheme

(a)	$A=\frac{3}{2}[0+0+2(15+18+15+12+15+18+15]$ $A=324\ m^2$	**Scale 15D (0, 4, 7, 11, 15)** *Low Partial Credit:* • Any correct dimension identified • Correct answer without work *Mid Partial Credit* • Correct formula with some correct substitution *High Partial Credit:* • Correct formula fully substituted correctly
(b)	$1{\cdot}6 \times 60 \times 60 \div 1000 = 5{\cdot}76$	**Scale 10C (0, 3, 7, 10)** *Low Partial Credit:* • $1{\cdot}6 \times 60$ or equivalent • Any work of merit *High Partial Credit:* • $1{\cdot}6 \times 60 \times 60 = 5760$ • $1{\cdot}6 \times 60 \times 60 \div 1000$ not finished

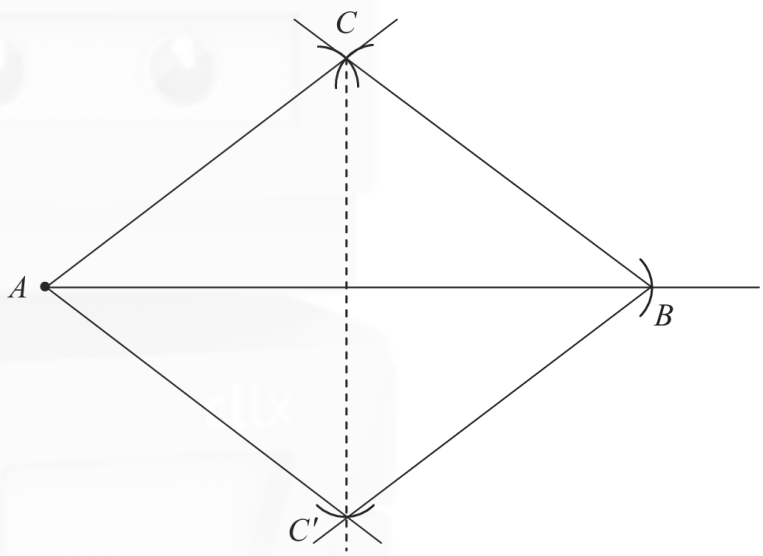
(a) Construct the triangle ABC such that $|AB| = 8$ cm, $|BC| = |AC| = 5$ cm. The point A is given to you.



(b) On the same diagram, construct the image of the triangle ABC under the axial symmetry in AB.

(c) Justify the statement " $AC'BC$ is a parallelogram" where C' is the image of C under the axial symmetry in AB.

(a) Construct the triangle ABC such that $|AB| = 8$ cm, $|BC| = |AC| = 5$ cm. The point A is given to you.



(b) On the same diagram, construct the image of the triangle ABC under the axial symmetry in AB.

(c) Justify the statement " $AC'BC$ is a parallelogram" where C' is the image of C under the axial symmetry in AB.

The diagonals $[AB]$ and $[CC']$ of $AC'BC$ bisect one another.

Hence, $AC'BC$ is a parallelogram.